HealthConnect — Domain Context

About the Organisation

HealthConnect is the digital health division of **Meridian Health Group**, a regional hospital network operating 3 acute-care hospitals and 12 outpatient clinics across the South-East. The group serves approximately 420,000 registered patients and processes 1.8 million appointment interactions per year.

Business Domain Concepts

User Personas

Persona	Description	Key Needs
Patient	Registered individuals receiving care	Self-service booking, records access, medication management
Carer / Guardian	Family members managing a dependant's care	Proxy access, appointment management on behalf of patient
Clinical Staff	Nurses, GPs, specialists	Inbox management, care plan updates, secure messaging
Front Desk / Admin	Clinic reception staff	Appointment queue, patient check-in, waitlist management
Clinical Informatics	Internal analytics team	Aggregate reporting, engagement metrics

Appointment Types

- **GP Appointment** — standard 10–15 min slot, bookable by patient up to 4 weeks ahead
- **Specialist Referral** — requires GP referral, booked on behalf of patient by admin
- **Diagnostic / Imaging** — radiology, pathology, booked directly after referral
- **Telehealth** — video consultation, requires video integration
- **Urgent / Walk-in** — same-day, queue-based, no advance booking

Medical Records & Data

- Patient records are the **source of truth in Epic EHR** (version Hyperdrive 2024)
- Meridian uses **HL7 FHIR R4** as the standard integration interface to Epic
- Lab results are delivered via **Epic → FHIR → Platform** within 4 hours of verification
- Radiology reports are delivered as PDF attachments via the same FHIR pipeline
- Patients may access records for the last **5 years** only (older records available on request)

Medication Management

- Prescriptions are issued and managed inside Epic
- Repeat prescription requests flow **from patient portal → GP inbox → Epic**
- Controlled substances cannot be requested via the portal (legislative constraint)
- Medication reminders are opt-in and must respect patient communication preferences

Secure Messaging

- All patient-to-clinician communication is **asynchronous** (not live chat)

- Response SLA is 2 business days for non-urgent messages
- Urgent messages trigger an escalation to the on-call nurse
- Messages become part of the clinical record (stored in Epic)
- Attachments allowed: photos (< 5 MB), PDFs (< 10 MB)

Compliance & Regulatory Context

HIPAA (Primary)

- All PHI (Protected Health Information) must be encrypted **at rest and in transit**
- Minimum Necessary Standard: patients only see their own data
- Audit logs required for every PHI access event (who, what, when)
- Patient has the right to access, correct, and download their records (Right of Access)
- Business Associate Agreements (BAA) required for all third-party processors

GDPR (Secondary — applies to patients in EU territories)

- Consent must be granular: separate opt-ins for marketing, research, and clinical communications
- Right to erasure: non-clinical data can be deleted on request
- Data residency: EU patient data must not leave EU data centres

Accessibility

- WCAG 2.1 AA compliance mandatory across all patient-facing interfaces
- Screen reader support required (JAWS and VoiceOver)
- Minimum font size 16px, 4.5:1 contrast ratio

Business Rules

1. A patient must verify their identity via their **date of birth + NHS/MRN number** before accessing records
2. Appointment cancellations within **2 hours of the slot** trigger a no-show flag in Epic
3. A patient may not hold more than **3 future bookings** for the same clinic at the same time
4. Repeat prescription requests must be submitted at least **7 days before** the medication runs out
5. Proxy access (carer/guardian) requires **written consent** from the patient, stored as a scanned document
6. Failed login attempts: account locked after **5 consecutive failures**, unlocked via email verification
7. Session timeout: **15 minutes** of inactivity for web, **30 minutes** for mobile (clinical staff: 8 minutes)
8. Telehealth sessions can only begin when **both patient and clinician** have joined the waiting room

Integration Landscape

System	Purpose	Protocol
Epic EHR	Source of truth for records, appointments, prescriptions	HL7 FHIR R4
Twilio	SMS and voice notifications	REST API
SendGrid	Transactional email	REST API
Doxy.me	Telehealth video sessions	WebRTC / iframe embed
Stripe	Co-payment collection (outpatient only)	REST API

Apple Health / Google Fit	Wearable data sync (Phase 2)	HealthKit / Health Connect